

Nyi Nyi Zin

Software Developer

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Profile

A self-taught Full-Stack Developer with a strong foundation in computer science principles, including advanced Algorithms and Data Structures, scalable software architecture, and modern design patterns. Experienced in building real-world applications through freelance projects and working with development teams, and leading projects through all phases of the software development lifecycle. Skilled in defining software architecture to be future-proof, designing database structures, and optimizing performance to ensure maintainable and scalable solutions. Passionate about continuous learning, solving complex problems, and delivering high-quality systems across both frontend and backend.

Education

Present	University of Computer (Sittwe)
Estimated Graduated 2027	computer science

Professional Experience

Nov 2025 - Present Yangon , Myanmar (Remote)	Software Development Team Leader (Full Time) waansaung.com  Led software development team, conducted code reviews, ensured code quality and documentation across frontend and backend, and promoted knowledge sharing and best practices. Defined project architecture and workflow, created flow diagrams, documented project processes, and implemented project-wide technical guidelines and standards. Assigned tasks, tracked project progress, and coordinated across frontend, backend, and UI/UX teams to ensure timely and high-quality deliverables. Led UI/UX team, provided guidance and feedback, ensuring design consistency and user-friendly interfaces. Managed and deployed full-stack infrastructure using Alibaba Cloud, including server setup, configuration, and performance monitoring. Implemented CI/CD pipelines with GitHub Actions to automate building, testing, and deployment for both frontend and backend applications, improving development efficiency and release reliability.
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April 2025 - Nov 2025 Yangon, Myanmar (Remote)	Fullstack Software Developer (Full Time) , triosys.info  TRIOSYS IT Solutions and Services Company, established in 2019, began by providing IT support to small businesses. Due to its strong focus on customer satisfaction, the company quickly built a positive reputation and subsequently registered officially under DICA. Today, TRIOSYS specializes in delivering innovative IT solutions and business services to a wide range of organizations. By leveraging cutting-edge technologies, the company helps clients streamline their operations, enhance efficiency, and achieve long-term, sustainable growth.
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At TRIOSYS, I was responsible for leading full-cycle software development projects, including front-end development, scalable back-end systems, and system architecture design. My key contributions included:

Frontend Development: Architected high-performance Next.js frontend using server components, dynamic imports, code splitting, and tree shaking, reducing initial loading. Implemented image optimization (WebP/AVIF, responsive, lazy loading) and CDN delivery for faster global performance. Developed Progressive Web App (PWA) with offline support, and push notifications. Optimized data-heavy workflows with virtualized lists, infinite scrolling, and debounced search. Built secure authentication system (JWT, refresh tokens, RBAC) and enforced frontend security via Zod validation, CSP, CSRF protection, and secure HTTP headers.

Streamlined form handling with React Hook Form + Zod, ensuring type-safe validation and minimal re-renders.

Introduced observability and reliability improvements: Sentry monitoring, error boundaries, retry logic, graceful fallback UIs, and performance monitoring using Lighthouse. Built automated testing suites with Jest and Vitest to ensure code quality and prevent regressions. Containerized frontend with Docker, integrated CI/CD pipelines (Jenkins), and applied Husky Git hooks for code quality enforcement.

Backend Development: Optimized database interactions with efficient queries, indexing, connection pooling, and sharding, reducing latency across Golang, Express, NestJS, and Next.js. Implemented caching strategies Redis, and HTTP caching to accelerate API responses and reduce server load. Enhanced API efficiency through pagination, bulk operations, payload minimization, and optimized data serialization. Scaled backend systems via horizontal scaling, load balancing, auto-scaling, and performance-tuned Golang/Node.js services. Enforced rate-limiting, input validation, and security best practices to maintain performance under high traffic.

May 2023 – Feb 2025
Thantwe, Myanmar

Associate Developer, DomiTech

DomiTech is a dynamic IT company specializing in both infrastructure solutions and software development. The company provides end-to-end technology services, including networking, system integration, and custom software applications. With a focus on innovation and efficiency, DomiTech delivers solutions that help businesses optimize operations, enhance digital capabilities, and stay ahead in a rapidly evolving technological landscape.

At DomiTech, I started with end-to-end networking services, managing infrastructure, connectivity, and system integration. Later, I transitioned into software development, contributing to the design and deployment of custom applications. This journey gave me a strong foundation in both IT infrastructure and software engineering, enabling me to deliver complete, end-to-end technology solutions.

Front-End Development: Worked on the development of complex UI components using React.js and TypeScript, focusing on performance, accessibility, and maintainable code architecture. Implemented client-side routing, form validation, and dynamic data fetchig.

Back-End Engineering: Designed and built RESTful APIs using Node.js and Express.js, handling features such as authentication (JWT, OAuth), file uploads, and role-based access control. Ensured API scalability and security.

Team Workflow: Collaborated in Git-based team environments, performed code reviews, and followed best practices in code structure, testing, and documentation.

Database Design & Integration: Modeled and optimized databases using MongoDB and PostgreSQL, wrote complex queries, and implemented data validation and indexing strategies for performance.

Skills

Frontend: HTML | CSS | JavaScript | TypeScript | React.js | Next.js | Tailwind CSS | Bootstrap | Shadcn UI | MUI (Material UI) | Mantine UI | Framer Motion

State & Data: Redux | Redux Toolkit | Zustand | TanStack Query (React Query) | TanStack Table

Forms & Validation: React Hook Form | Formik | Zod

Mobile: Flutter | Dart | React Native | Expo | React Native CLI

Backend: Node.js | Express.js | NestJS | Golang (Gin | Fiber | Echo)

Databases: MySQL | MSSQL | MongoDB | PostgreSQL | MariaDB | NeonDB | Firebase | Firestore

ORM: Prisma ORM | Drizzle ORM | Sequelize ORM | Mongoose ORM | GORM

API & Communication: REST API | GraphQL | gRPC | WebSocket

Authentication & Security: JWT | OAuth | bcrypt

DevOps & Tools: Git | Linux | Docker | Alibaba Cloud | Jenkins | GitHub Actions | CI/CD pipelines | Nginx | OpenResty | Husky | ESLint | Prettier | Postman | Insomnia

Architecture: MVC Pattern | Clean Architecture | Microservices | Layered Architecture | Monolithic | Modular Monolithic | Event-Driven

Caching & Performance: Redis | CDN | Browser caching | HTTP caching | Lazy loading | Code splitting | Debouncing / Throttling | Tree shaking | Image optimization | Indexing | Rate limiting | Background jobs / queues | Core Web Vitals optimization (LCP, FID, CLS)

Monitoring: Sentry | Datadog

Other: PWA (Progressive Web Apps)

Featured Projects

Price Change System

Full-stack application for managing and approving product pricing across multiple sales channels with structured approval workflows, audit tracking, and multi-currency support.

Key Contributions:

- Built end-to-end multi-channel price management workflow (create, submit, approve/reject, track status)
- Implemented role-based approval system for price change requests with configurable permissions
- Designed and implemented RBAC for fine-grained access control and super admin management
- Developed secure authentication using JWT with refresh token support
- Integrated audit logging and historical tracking for all price changes
- Implemented multi-currency pricing with automatic exchange rate conversion
- Built inventory management features including barcode scanning and product catalog handling
- Enabled Excel import/export for bulk pricing and product data operations
- Designed responsive UI with dashboards, tables, and workflow-driven forms

Tech Stack: NestJS, React (Vite), PostgreSQL (Prisma), Docker, Mantine UI, Tailwind CSS, Redux Toolkit, React Query, TypeScript, Jenkins

Expense Request Management System

Full-stack application for managing employee expense requests with approval workflows, real-time updates, and financial data integration.

Key Contributions:

- Built end-to-end expense request workflow (submit, approve/reject, track status)
- Implemented multi-level approval system with configurable policies
- Designed and implemented RBAC + ABAC for fine-grained access control
- Developed secure authentication using JWT and protected APIs
- Integrated real-time updates using WebSocket and Firebase notifications
- Managed master data (categories, projects, GL accounts, payment methods)
- Enabled Excel export and attachment handling for expense records
- Designed responsive UI with forms, tables, and dashboards

Tech Stack: Go (Echo), React (Vite), MySQL (GORM), Docker, Mantine, Tailwind CSS, React Query, TypeScript, Firebase

Wedding Invitation Website

Full-stack digital wedding invitation platform where couples can share a website link instead of traditional cards, featuring interactive guest engagement and real-time event updates.

Key Contributions:

- Built digital wedding invitation system with couple profiles, photo showcase, and event sharing via link
- Implemented live wish/comment feature allowing guests to send real-time messages
- Developed real-time wedding schedule display for event timeline visibility
- Enabled "Add to Calendar" functionality for saving wedding date and time directly to mobile devices
- Integrated real-time updates for guest interactions and event content
- Designed responsive and visually rich UI for invitation experience and guest engagement

Tech Stack: Next.js, TypeScript, Shadcn UI, Go (Echo), GORM, PostgreSQL

Educational Platform jca.com.mm

Full-stack multi-application web system for managing and delivering educational content for a service academy (JCA), including admin management, public user portal, and backend API services.

Key Contributions:

Built full-stack multi-app architecture separating backend API, admin dashboard, and public user platform

Implemented Go-based REST API with Echo framework for routing, middleware, validation, and authentication

Designed and developed modular backend system with controllers for courses, articles, announcements, users, roles, and more

Integrated JWT-based authentication with role/permission management and secure API access

Implemented background job processing system for asynchronous tasks (queues/worker architecture)

Built file upload and static asset management system with server-side handling

Developed admin dashboard for full content management (courses, users, articles, announcements, etc.) using React, Vite, TypeScript, and React Query

Implemented rich text editing, data visualization dashboards, and state management using Zustand

Built public-facing website using Next.js with dynamic content rendering and SEO-friendly structure

Integrated modern UI/UX components with Tailwind CSS, Radix UI, animations, and media handling libraries

Ensured clean developer experience with linting, formatting, and CI tooling (ESLint, Prettier, Husky)

Tech Stack: Go (Echo, GORM, PostgreSQL), Next.js, React (Vite), TypeScript, Tailwind CSS, React Query, Zustand, Radix UI, TipTap, Recharts, JWT, AWS SDK, Docker-ready structure

Company Profile Website triosys.info

Modern company profile website for Triosys, redesigned as a high-performance, SEO-friendly web application to showcase IT services, team expertise, certifications, and contact information.

Key Contributions:

- Built responsive multi-section company website using Next.js with server-side rendering for improved performance and SEO
- Implemented modular component architecture using Shadcn UI for consistent and reusable design system
- Developed structured service showcase covering IT solutions, network security, website development, outsourcing, and support services
- Designed dynamic content sections for team profiles, certifications, and company portfolio presentation
- Integrated optimized asset management system for images, certificates, and media resources
- Implemented smooth navigation and section-based UI with modern UX interactions and animations
- Built fully responsive interface using Tailwind CSS with mobile-first design approach
- Improved performance and scalability through Next.js routing, layout optimization, and reusable components

Tech Stack: Next.js, TypeScript, Shadcn UI, Tailwind CSS

AI Eye Tracking Focus Monitoring System

Real-time AI-based webcam system that tracks user face and eye movements to monitor focus and detect distraction or drowsiness, helping improve productivity during computer usage.

Key Contributions:

Built real-time face and eye tracking system using OpenCV and MediaPipe
Implemented gaze and head position detection to monitor user attention
Developed eye closure detection (EAR-based) for drowsiness monitoring
Integrated alert system with sound notifications for distraction and prolonged eye closure
Designed real-time video processing pipeline for webcam-based analysis
Implemented focus classification logic using head pose, iris movement, and eye state

Tech Stack: Python, OpenCV, MediaPipe, NumPy, Winsound

Task Manager Platform

Full-stack task management system with a modern SPA frontend, backend API services, and real-time collaboration features for efficient task tracking and team coordination.

Key Contributions:

- Built single-page application (SPA) frontend served via Nginx with client-side routing and fallback handling
- Implemented backend API integration for task creation, updates, deletion, and management workflows
- Integrated real-time communication system using Socket.IO for live updates and notifications
- Designed reverse proxy configuration in Nginx to handle API routing, WebSocket connections, and static file serving
- Implemented file upload and direct static file serving for attachments and shared resources
- Configured Docker-based service communication between frontend, backend, and real-time services
- Optimized static asset caching and routing for improved performance and scalability
- Structured system for modular separation of frontend, backend, and real-time services

Tech Stack: React (SPA), Nginx, Node.js (API service), Socket.IO, Docker, WebSocket, Static File Serving

Face Recognition System

Face Recognition Security Monitoring System, a real-time computer vision application that detects and verifies authorized individuals using webcam input for basic surveillance and access control.

Key Contributions:

- Built real-time face detection and recognition using OpenCV and face_recognition
- Implemented dataset-based authorized face verification system
- Developed facial encoding and matching logic with configurable accuracy
- Added visual indicators for recognized and unrecognized faces
- Handled live webcam streaming and frame processing efficiently

Tech Stack: Python, OpenCV, face_recognition, NumPy

Delivery System (Super App) waansaung.com

WannSann Delivery Job Marketplace, a scalable super app connecting delivery workers and businesses through a two-sided marketplace with real-time job matching, order management, and multi-platform support.

Key Contributions:

- Built multi-platform super app (Web, Mobile, Admin) using Next.js, React Native, and Expo
- Developed backend services with NestJS, PostgreSQL, and Prisma for job, order, and user management
- Implemented role-based access control with JWT and Google OAuth authentication
- Designed two-sided marketplace for employer job posts and worker seeker posts
- Integrated internationalization (EN, MM, TH, CN) and rich text job descriptions
- Optimized performance with Redis caching, lazy loading, and indexed queries
- Deployed scalable, Dockerized system on Alibaba Cloud with OSS integration for file storage
- Maintained modular and extensible architecture for long-term scalability

Tech Stack: Next.js, React Native, Expo, NestJS, PostgreSQL, Prisma, Redis, TypeScript, Alibaba Cloud

Face Mask Detection System,

a real-time computer vision application that detects whether individuals are wearing face masks using image or video input for public safety monitoring and compliance.

Key Contributions:

- Built real-time face mask detection using OpenCV and deep learning models
- Trained a custom model on Google Colab for accurate mask classification
- Implemented image and video stream processing for live detection
- Developed classification logic to distinguish masked vs unmasked faces
- Optimized detection performance for real-world scenarios such as CCTV monitoring

Tech Stack: Python, TensorFlow, PyTorch, Keras, OpenCV, NumPy

Certificates

Web Development Course - MERN (**Code Hub**)

Next.js Mastery Course (**Code Hub**)

Docker Course (**Code Hub**)

Go:The Complete Developer's Guide(Golang) (**Udemy**)

My Portfolio Website

<https://nyinyizin-portfolio.vercel.app/>

Learn more about me